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Try to answer the following questions:

Q1: What do we mean by a drug expiry date? Does it differ if the formulation is used or not? What is meant by shelf-life of a drug?

Q2: What are the differences between trade and generic names of a drug?

Q3: Are all drug tablets breakable into two halves?

Q4: What are the differences between extended (slow), sustained release, controlled-release, modified-release, ...?

Q5: What are the different coats of a tablet form of a drug?

Q6: Can you chew an enteric-coated (gastric-resistant) tablet before swallowing? Give one example of such formulation.

Q7: What is the difference between pessaries and suppositories?

Q8: What are the differences between an ointment, cream, gel, and lotion?

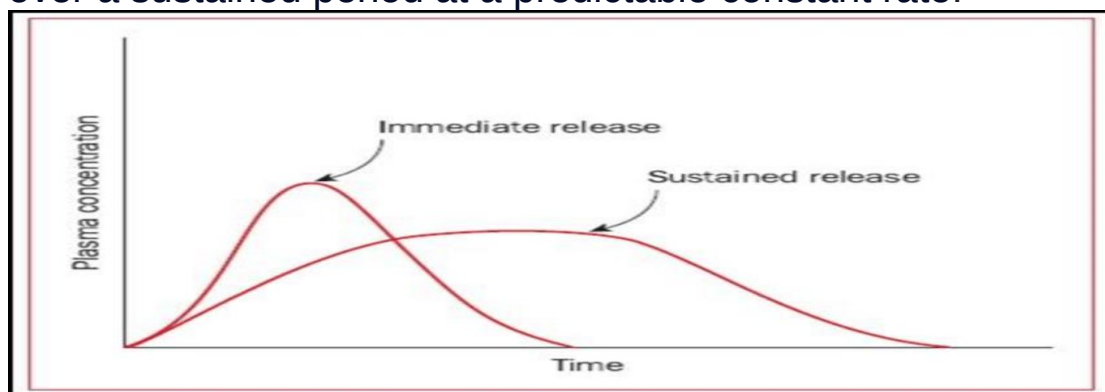
Q9: How does an ampule differ from a vial?

Q10: How would you break an ampule?

Q11: Why the 500 mg paracetamol tablet weighs 600-650mg?

Q12: What is the difference between a syrup and an elixir?

Extended-release dosage consists of either **sustained**-release (SR) or **controlled**-release (CR) dosage. SR maintains drug release over a sustained period but at nearly constant rate (but not predictable). CR maintains drug release over a sustained period at a predictable constant rate.



Q1: What do we mean by a drug expiry date? Does it differ if the formulation is used or not? What is meant by shelf-life of a drug?

- **Expiry date:** The date until which the manufacturer guarantees the drug's full potency and safety, if unopened and stored correctly (What else does it mean?)
 - **Opened formulation:** Once opened (e.g., eye drops, insulin vials), stability may be reduced, and they must be discarded earlier (e.g., 28 days for many eye drops).
 - **Shelf-life:** The period during which the drug remains (on the shelf) stable, safe, and effective under recommended storage conditions (i.e. time between manufacturing date and expiry date).
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Q2: What are the differences between trade and generic names of a drug?

- **Generic name:** The international non-proprietary name (e.g., *paracetamol*). Same worldwide.
- **Trade (brand) name:** A name given by the manufacturer for marketing purposes (e.g., *Panadol*, *Tylenol*).

Brand (Trade, Proprietary) name	Generic (Official, Scientific, Non-proprietary) name
Used for marketing	Used for scientific description of the API (shortened chemical name)
Given by the pharmaceutical company that manufactures it	Given by official bodies such as WHO, USAN,...
Carries the symbol TM or [®] with (Panadol)	Written in small letters (Lowercase)(paracetamol)
Short and easy to remember (Voltaren)	Often long and based on the chemical name (diclofenac sodium)
Multiple names for one drug (different companies have different brand names)	One only for each drug (API)
Owned by the company (proprietary)	Public property (can be used by anyone)
Expensive	Lower cost

Q3: Are all drug tablets breakable into two halves?

- No. Only **scored tablets** (with a line) are designed to be split.
 - **Unscored, coated, or modified-release tablets** must not be broken as it can alter absorption and safety.
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Q4: Differences between extended, sustained release, controlled-release, modified-release, ...?

- **Extended/Prolonged release:** General terms for dosage forms releasing drug slowly over time.
 - **Sustained release:** Maintains drug release at a nearly constant rate for a prolonged period.
 - **Controlled release:** More precise; delivers drug at a predictable, controlled rate.
 - **Modified release:** Broad term that includes any formulation altering the time/place/rate of release (e.g., delayed release, extended release).
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Q5: What are the different coats of a tablet form of a drug?

- **Sugar coat** – for taste masking, appearance.
 - **Film coat** – thin polymer layer, improves swallowing, stability.
 - **Enteric coat** – resists stomach acid, dissolves in intestine.
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Q6: Can you chew an enteric-coated (gastric-resistant) tablet before swallowing? Give one example.

- **No.** Chewing destroys the protective coating, leading to drug inactivation or stomach irritation.
 - **Example:** Enteric-coated *aspirin* tablets.
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Q7: What is the difference between pessaries and suppositories?

- **Pessaries:** Solid dosage forms inserted into the **vagina** (e.g., antifungal pessaries for candidiasis).
 - **Suppositories:** Solid dosage forms inserted into the **rectum** (e.g., glycerin suppositories for constipation).
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Q8: Differences between ointment, cream, gel, and lotion?

- **Ointment:** Oil-based, greasy, occlusive, good for dry skin.
 - **Cream:** Water + oil emulsion, less greasy, good for moist lesions.
 - **Gel:** Water/alcohol base, transparent, cooling, quick absorption.
 - **Lotion:** Liquid, easy to apply over large/haired areas.
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Q9: How does an ampule differ from a vial?

- **Ampule:** Sealed glass container (single dose); must be broken to open.
 - **Vial:** Glass/plastic container sealed with rubber stopper; can be **single or multi-dose**.
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Q10: How would you break an ampule?

1. Tap the top to remove liquid from neck.
 2. Use an alcohol swab or gauze to wrap neck.
 3. Snap the neck **away from your body** at the scored line or at the site of the marked dot.
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Q11: Why does a 500 mg paracetamol tablet weigh 600–650 mg?

- Because tablets contain **excipients** (binders, fillers, disintegrants, lubricants, coatings) in addition to the **active drug**.
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Q12: What is the difference between a syrup and an elixir?

- **Syrup:** Sweet, viscous, water-based solution, often for children, no (or very little) alcohol.
 - **Elixir:** Clear, hydro-alcoholic solution, less viscous, better solvent for some drugs.
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A concise summary table of the Q&As for quick revision:

Q	Answer (Summary)
Q1: Expiry date / Shelf-life	Expiry = date of guaranteed potency if unopened. Shelf-life = duration of stability. Opened products (e.g., eye drops) expire sooner.
Q2: Trade vs Generic name	Generic = official drug name (<i>paracetamol</i>). Trade = brand name (<i>Panadol</i>).
Q3: Tablet splitting	Only scored tablets can be split. Coated/modified-release must not be broken.
Q4: Release types	Extended = slow release. Sustained = constant rate. Controlled = predictable release. Modified = any altered release (delayed, extended, etc.).
Q5: Tablet coats	Sugar coat (taste), Film coat (swallowing/stability), Enteric coat (acid-resistant),
Q6: Enteric-coated tablets	Must not be chewed (coating destroyed). Example: Enteric-coated aspirin.
Q7: Pessary vs Suppository	Pessary = vaginal use. Suppository = rectal use.
Q8: Ointment/Cream/Gel/Lotion	Ointment = greasy, dry skin. Cream = less greasy, moist lesions. Gel = cooling, fast absorption. Lotion = liquid, for

Q	Answer (Summary)
	large/haired areas.
Q9: Ampule vs Vial	Ampule = sealed glass, single use. Vial = rubber stopper, single or multi-use.
Q10: Breaking ampule	Tap liquid down → wrap neck → snap away at scored line.
Q11: Paracetamol 500 mg ≠ Tablet weight	Extra weight from excipients (fillers, binders, coatings, etc.).
Q12: Syrup vs Elixir	Syrup = sweet, water-based, viscous, kids-friendly. Elixir = hydro-alcoholic, clear, less viscous.

Practical No. 2: Routes of Drug Administration
 Through group discussion, try to answer the following questions:

Route	Description
How would you dissolve a powder for oral administration?	
What is the proper method for using an inhaler?	
What is a spacer? Why is it, sometimes, used with the inhaler?	
What is a nebulizer? For what is it used?	
How would you break an ampoule?	
How would you dissolve a drug powder in a vial?	
How would you administer an irritant drug orally?	
How would you administer an irritant drug intramuscularly?	
What is a depot preparation? What are its properties?	

How would you store a suppository at home?	
How would you administer insulin to a patient? Can the patient inject him/herself?	
How would you administer drops (eye, nasal, ear)?	

1. How would you dissolve a powder for oral administration?

- Use the **recommended solvent** (usually clean water, preferably boiled and cooled, sometimes milk/juice if permitted).
 - Add solvent gradually while stirring/shaking until the powder is fully dissolved or suspended.
 - Use the solution immediately and keep in the fridge (many reconstituted powders are unstable).
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2. What is the proper method for using an inhaler?

1. Shake the inhaler well.
 2. Exhale fully.
 3. Place mouthpiece in mouth, lips sealed.
 4. Press canister **once** while breathing in slowly and deeply.
 5. Hold breath for ~10 seconds.
 6. Exhale slowly.
- (Rinse mouth after steroid inhalers to prevent fungal infection).*
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3. What is a spacer? Why is it, sometimes, used with the inhaler?

- **Spacer:** A tube/chamber attached to an inhaler.
 - **Purpose:**
 - Improves delivery of drug to lungs.
 - Reduces drug deposition in mouth/throat.
 - Helps patients (children, elderly) who cannot coordinate pressing + inhaling simultaneously.
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4. What is a nebulizer? For what is it used?

- **Nebulizer:** Device that converts liquid medication into a fine mist for inhalation via mask or mouthpiece.
 - **Uses:** For patients unable to use inhalers effectively (children, severe asthma/COPD attacks, hospitalized patients).
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5. How would you break an ampoule?

1. Tap the top to move liquid down.
 2. Wrap neck with gauze/swab.
 3. Snap the neck **away from body** at the scored mark or at the site of the dot.
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6. How would you dissolve a drug powder in a vial?

1. Clean rubber stopper with alcohol swab.
 2. Withdraw diluent (sterile water or saline) into syringe.
 3. Inject diluent into vial slowly.
 4. Swirl gently (do not shake vigorously unless instructed).
 5. Withdraw reconstituted solution for use.
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7. How would you administer an irritant drug orally?

- Give **after meals** to minimize stomach irritation.
 - Use with plenty of water or milk (if appropriate).
 - Some may require **enteric coating** to avoid gastric irritation.
 - Patient should sit or stand upright before and after swallowing.
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8. How would you administer an irritant drug intramuscularly?

- Inject **deep IM** into a large muscle (gluteus, vastus lateralis).
 - Use **Z-track technique** (pull skin sideways before injection, release after) to prevent leakage into subcutaneous tissue and irritation.
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9. What is a depot preparation? What are its properties?

- **Depot preparation:** A drug formulation designed for slow release over a prolonged period (days–weeks).
 - **Properties:**
 - Injected IM or SC.
 - Slow absorption.
 - Long duration of action.
 - Reduces frequency of dosing (e.g., antipsychotics, contraceptives).
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10. How would you store a suppository at home?

- Store in a **cool place or refrigerator** (not freezer).
 - The tip of the suppository should be kept down.
 - Protect from heat to avoid melting.
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11. How would you administer insulin to a patient? Can the patient inject him/herself?

- **Administration:** Subcutaneous injection (abdomen, thigh, upper arm). Rotate injection sites.
 - **Yes**, with proper training, most patients can self-inject using insulin syringes or pens.
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12. How would you administer drops (eye, nasal, ear)?

- **Eye drops:** Tilt head back, pull lower eyelid, instill drop, close eye gently, avoid touching dropper tip.
- **Nasal drops/spray:** Tilt head back slightly, instill drops into nostril, sniff gently.
- **Ear drops:**
 - Adults: Pull pinna **up and back**.
 - Children: Pull pinna **down and back**.
 - Instill drops, keep head tilted for a few minutes.